

Lecture 7 - Part III

Consumer Choice

ECON 101
Fall 2020



The Consumer Problem

Choose the best bundle, according to the consumer's preferences,
subject to the consumer's budget constraint.

"Choose the best among what is feasible"



$$\max_{(x_1, x_2)} U(x_1, x_2)$$

$$\text{subject to: } \underline{p_1 x_1 + p_2 x_2 \leq I}$$

p_1 : price of good 1

p_2 : price of good 2

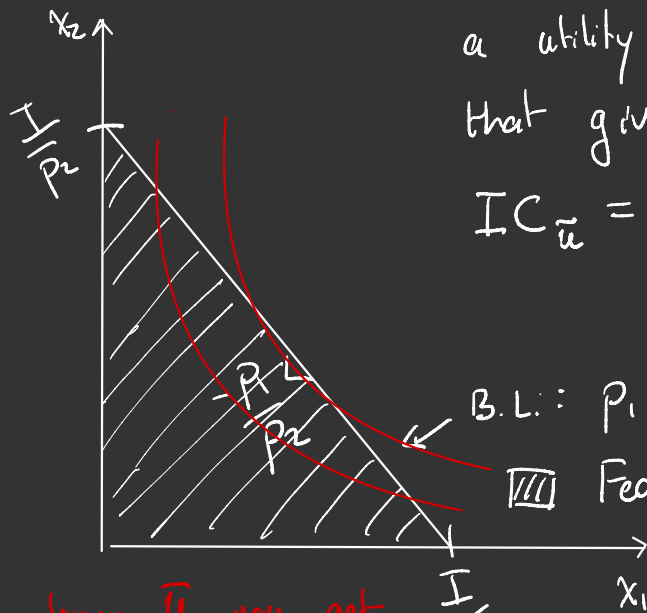
I : income

The feasible set of the
consumer

Recall that an indifference curve, given a utility function, is the set of points that give equal utility

$$IC_{\bar{u}} = \{(x_1, x_2) \mid \bar{u} = u(x_1, x_2)\}$$

We can compute the optimal (best) consumption bundle for the consumer.



$$B.L.: p_1 x_1 + p_2 x_2 = I; \quad x_2 = \frac{I}{p_2} - \frac{p_1}{p_2} x_1$$

Feasible/Budget Set

For a larger $\bar{u}$, you get a higher indifference curve.

Example:

$$u(x_1, x_2) = x_1 x_2 \Rightarrow IC_{\bar{u}}:$$

Eq. of this indifference curve is

$$\bar{u} = x_1 x_2$$

$$x_2 = \frac{\bar{u}}{x_1}$$

Optimal Bundle (market basket)

If preferences are:

1. Complete
2. Transitive
3. More is better

} Optimal
Bundle

$$A \sim B \sim C$$

1. On the budget line
2. On the highest possible I.C.

} I.C. of the consumer must be tangent to the B.L. at an optimum.

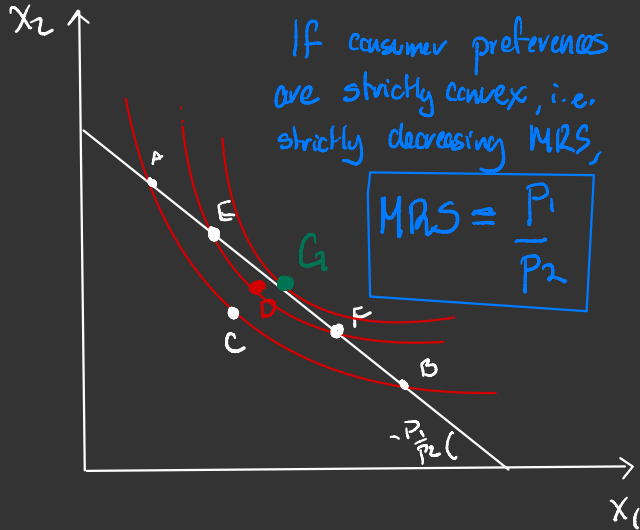
A, B are not optimal: why? because we can do better.

C is not optimal: not on the budget line

D is not optimal: not on the budget line

E, F are not optimal: because we can do better

G is optimal: is in the budget line and we cannot put the consumer in a higher I.C. "the consumer cannot achieve a higher I.C., i.e. a better consumption bundle"



Consumer Problem: $\max_{(x_1, x_2)} U(x_1, x_2)$
s.t. $p_1 x_1 + p_2 x_2 \leq I$

If preferences are

1: Complete

2: Transitive

3: More is better

(3) allows me to focus on bundles exactly on the B.L. $\rightarrow$

$$\max_{(x_1, x_2)} U(x_1, x_2)$$

s.t. $p_1 x_1 + p_2 x_2 = I$

$\hookrightarrow$ Consumer will exhaust his whole budget

+ 4. Strict Convexity (MRS strictly decreasing)

"strict preference for balancedness"

$$\max_{(x_1, x_2)} U(x_1, x_2)$$

s.t. $p_1 x_1 + p_2 x_2 = I$

optimality
 $\longrightarrow$
condition

$$\text{MRS} = \frac{p_1}{p_2} \quad \left| \begin{array}{l} \text{1 system} \\ \text{2 eqs.} \\ \text{in} \\ \text{2 unknowns} \end{array} \right.$$

$$p_1 x_1 + p_2 x_2 = I$$

Example: $U(x_1, x_2) = x_1 x_2$

Consumer problem is: $\max_{(x_1, x_2)} x_1 x_2$
s.t. $p_1 x_1 + p_2 x_2 = I$

One way to solve: $x_2 = \frac{I}{p_2} - \frac{p_1}{p_2} x_1$, plug into the utility function:

$\max_{x_1} x_1 \left(\frac{I}{p_2} - \frac{p_1}{p_2} x_1 \right) \rightarrow$ one-variable maximization problem
 $\hookrightarrow$ You can solve this with calculus.

MRS for $U(x_1, x_2) = x_1 x_2$ is given by $MRS(x_1, x_2) = \frac{x_2}{x_1}$

Optimality condition is

$MRS = \frac{p_1}{p_2} \Leftrightarrow \frac{x_2}{x_1} = \frac{p_1}{p_2}$, together $p_1 x_1 + p_2 x_2 = I$, identify the optimal bundle.

Working with the same example, assume that

$$P_1 = 1, P_2 = 2, I = 80$$

$$\max_{(x_1, x_2)} x_1 x_2$$

$$\text{s.t. } (1)x_1 + (2)x_2 = 80$$

Opt. Condition

$$MRS \frac{x_2}{x_1} = \frac{1}{2} = \frac{P_1}{P_2}, \quad x_1 + 2x_2 = 80, \quad \text{identify the solution.}$$

$$\frac{x_2}{x_1} = \frac{1}{2} \Rightarrow x_2 = \frac{1}{2} x_1 \Rightarrow$$

plug this into
eq. B.L.

$$x_1 + 2\left(\frac{1}{2} x_1\right) = 80$$

$$2x_1 = 80 \Rightarrow$$

$$x_1 = 40$$

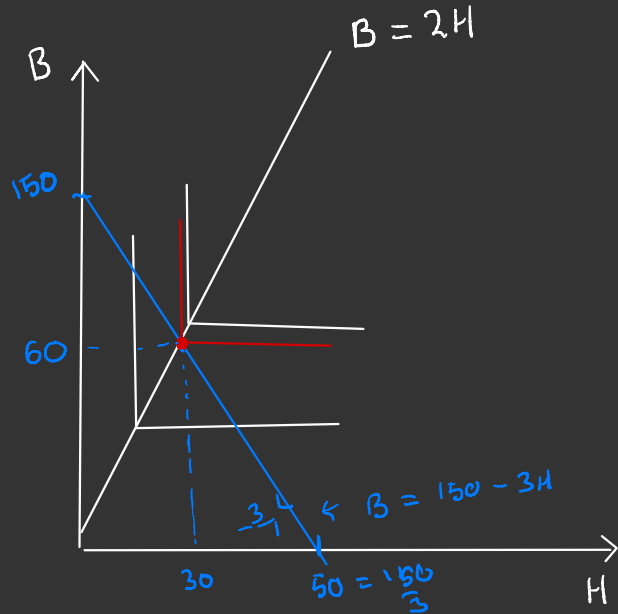
$$x_2 = 20$$

Special Cases:

Perfect Complements:

↳ IC are right angles

↳ MRS is not defined



$U(x_1, x_2) = \min \{ \alpha x_1, \beta x_2 \}$, Kink $\beta x_2 = \alpha x_1$

Sandwiches example: $\rightarrow$ 2 Slices of Bread
 $\rightarrow$ 1 Slice of Ham

$$U(H, B) = \min \{ 2H, B \}$$

$$\text{Kink} \Rightarrow 2H = B$$

↓ This preferences satisfy

1. Completeness
2. Transitivity
3. More is better

However, I don't even have a way to define the MRS!

Consumer's problem

$$\max_{(H, B)} \min \{ 2H, B \}$$

$$\text{s.t. } (1) B + (3) H = 150$$

Optimality condition:

$$2H = B \quad (\text{Kink equation})$$

$$\begin{aligned} &\& \\ B + 3H &= 150 \end{aligned}$$

$$P_H = 3, P_B = 1, I = 150$$

Plug in Kink eq. into BL

$$2H + 3H = 150$$

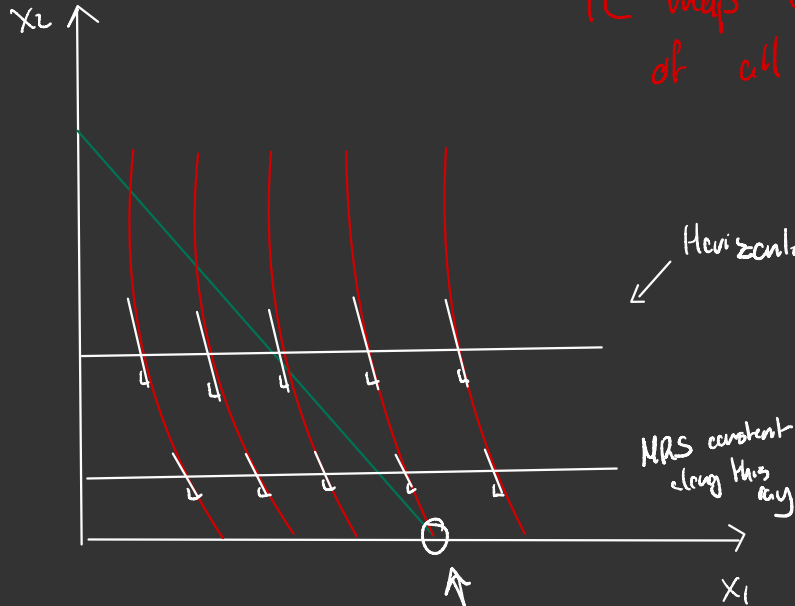
$$H = \frac{150}{5} = 30, B = 60$$

Special Case: Corner Solutions

- Usually a failure of Strict Convexity (but not all of the time)

B.L. —

IC map consists of parallel displacements of all the individual indifference curves



Horizontal Ray: The MRS is constant along this ray, i.e., the IC's have the same slope

MRS constant along this ray

This is the optimum

You will analyze this case in Intermediate Micro

Those preferences satisfy:

- 1: Completeness
- 2: Transitivity
- 3: More is better

4: Convexity (but not strict)

⊕ 5: MRS is constant along horizontal rays.

Corner Solution: Perfect Substitutes, IC's are straight lines with MRS constant along the IC.

General Case $U(x_1, x_2) = a x_1 + b x_2$; $MRS = \frac{a}{b}$;

$x_1 = \text{Coke}$ } substitutes
 $x_2 = \text{Pepsi}$ } but
 I slightly prefer Coke $\Rightarrow U(x_1, x_2) = 2x_1 + x_2$; $\bar{U} = 2x_1 + x_2$

$\Rightarrow$ IC eq: $x_2 = \bar{U} - 2x_1$; $MRS = 2$,

IC. —

my income is 500 Tenge

$p_1 = 150$ price of coke

$p_2 = 150$ price of pepsi

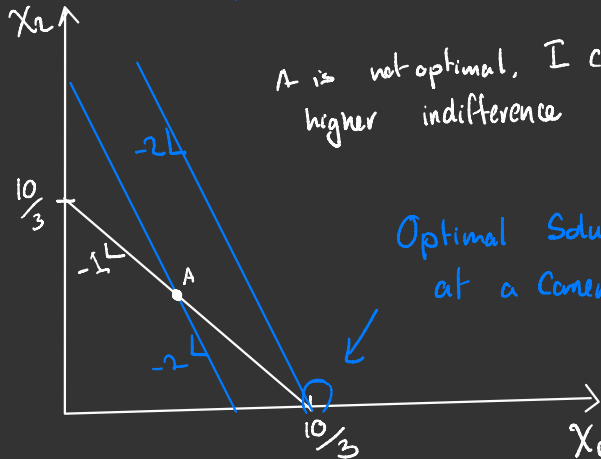
$\frac{p_1}{p_2} = \frac{150}{150} = 1$; Eq. BL $x_2 = \frac{500}{150} - x_1$

Eq: $x_2 = \frac{10}{3} - x_1$

A is not optimal, I can achieve a higher indifference curve

Optimal Solution is at a corner $\rightarrow$

$MRS = 2 > 1 = \frac{p_1}{p_2}$; I spend all money in coke I spend 500 tenge in coke



Suppose that coke is more expensive,

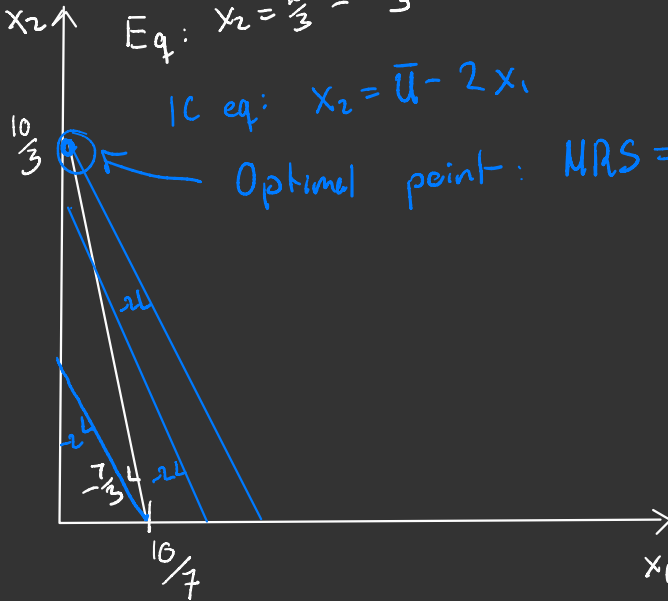
$$\left. \begin{array}{l} p_1 = 350 \text{ price of coke} \\ p_2 = 150 \text{ price of pepsi} \end{array} \right\} \frac{p_1}{p_2} = \frac{350}{150} = \frac{7}{3} = 2.\overline{33}$$

$$I = 500$$

Budget Line
Eq: $x_2 = \frac{10}{3} - \frac{7}{3}x_1$

IC eq: $x_2 = \bar{u} - 2x_1$

Optimal point: $MRS = 2 < \frac{7}{3} = \frac{p_1}{p_2}$



The optimality conditions for this example are as follows

$$MRS = 2;$$

• If $\frac{p_1}{p_2} < 2$ then $x_1 = \frac{I}{p_1}$
 $x_2 = 0$

• If $\frac{p_1}{p_2} > 2$ then $x_1 = 0$
 $x_2 = \frac{I}{p_2}$

• If $\frac{p_1}{p_2} = 2$ then $x_1 \in [0, \frac{I}{p_1}]$
 $x_2 = \frac{I}{p_2} - \frac{p_1}{p_2} x_1$

$$\max_{(x_1, x_2)} U(x_1, x_2) = 2x_1 + x_2$$

$$\text{s.t. } p_1 x_1 + p_2 x_2 = I$$

Why equal?

1. Complete
2. Transitive
3. More is Better

MRS = 2 in this example.

general

$$U(x_1, x_2) = ax_1 + bx_2$$

→
solution

$$MRS = \frac{a}{b}$$

$$\text{If } MRS > \frac{p_1}{p_2}, \quad x_1 = \frac{I}{p_1}$$

$$x_2 = 0$$

$$\text{If } MRS < \frac{p_1}{p_2}, \quad x_1 = 0$$

$$x_2 = \frac{I}{p_2}$$

$$\text{If } MRS = \frac{p_1}{p_2}, \quad x_1 \in [0, \frac{I}{p_1}]$$

$$x_2 = \frac{I}{p_2} - \frac{p_1}{p_2} x_1$$

